

Sec 6.1 – Antiderivatives Graphically and Numerically

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

Partner up

Fill in as much of the following chart as you can, reminding yourself what you know about f and f'' given information about f' . (Sometimes you may not know anything!)

f	f'	f''
	positive	
	negative	
	0	
	increasing	
	decreasing	

=====

f	f'	f''
increasing	positive	?  or 
decreasing	negative	?  or 
critical points	0 or DNE	?
concave up	increasing	positive
concave down	decreasing	negative

Definition

If the derivative of F is f , we call F an **antiderivative** of f .

Example: Since the derivative of $\sin(x)$ is $\cos(x)$, $\sin(x)$ is an antiderivative of $\cos(x)$.

Question: What is *another* antiderivative of $\cos(x)$? And *another*?
(Can you find *all* antiderivatives?)

Theorem

Suppose $F(x)$ is an antiderivative of $f(x)$. Then $F(x) + C$ is also an antiderivative of $f(x)$.

Recall from the Fundamental Theorem of Calculus that if F is an antiderivative of f that

$$F(b) - F(a) = \int_a^b f(x) dx.$$

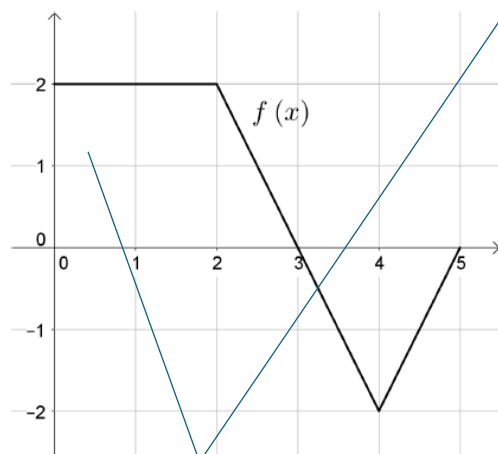
Then

$$F(b) = F(a) + \int_a^b f(x) dx.$$

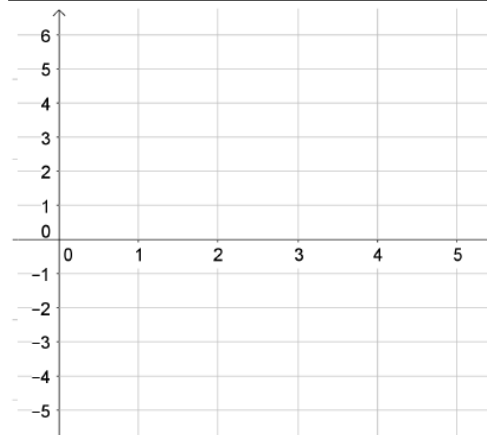
This means if we know $F(a)$ and $f(x)$, we can find values of our antiderivative by using definite integrals! The area we accumulate is how much $F(x)$ changes from $x = a$ to $x = b$.

Partner up

Suppose F is an antiderivative of f and $F(0) = 1$. Use the graph of $f(x)$ and your knowledge about integrals to fill in the table of values for $F(x)$ and then sketch the graph of $F(x)$ on the axes to the right.



x	0	1	2	3	4	5
$F(x)$						

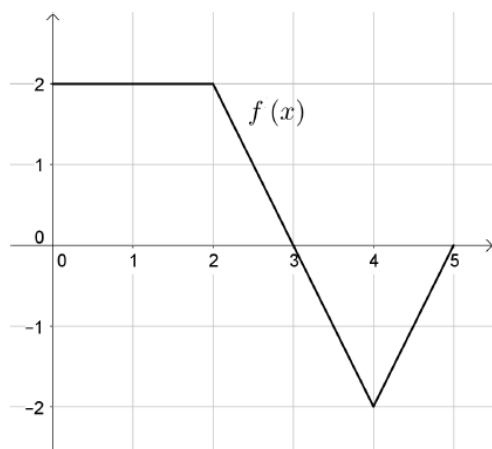


$$F(1) = F(0) + \int_0^1 f(x) dx, \quad F(1) = 1 + 2 = 3 \quad \text{Remember, } F'(x) = f(x)$$

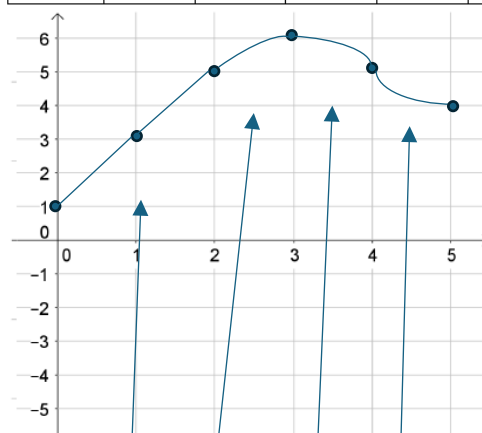
$$F(2) = F(1) + \int_1^2 f(x) dx \quad F(3) = F(2) + \int_2^3 f(x) dx \quad F(4) = F(3) + \int_3^4 f(x) dx \quad F(5) = F(4) + \int_4^5 f(x) dx$$

$$F(2) = 3 + 2 = 5 \quad F(3) = 5 + 1 = 6 \quad F(4) = 6 - 1 = 5 \quad F(5) = 5 - 1 = 4$$

Suppose F is an antiderivative of f and $F(0) = 1$. Use the graph of $f(x)$ and your knowledge about integrals to fill in the table of values for $F(x)$ and then sketch the graph of $F(x)$ on the axes to the right.



x	0	1	2	3	4	5
$F(x)$	1	3	5	6		



F increasing, constant slope

F increasing, c-down

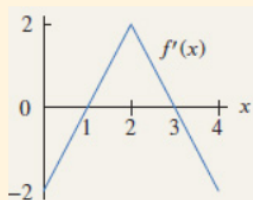
F decreasing, c-down

F decreasing, c-up

f	f'
increasing	positive
decreasing	negative
critical points	0 or DNE
concave up	increasing
concave down	decreasing

Time permitting, here is another example of how the graph of the derivative relates to the graph of the original function.

For the function f' given in [Figure 6.5](#), sketch a graph of three antiderivative functions f , one with $f(0) = 0$, one with $f(0) = 1$, and one with $f(0) = 2$.



[Figure 6.5](#) Slope function, f'

Solution

To graph f , start at the point on the vertical axis specified by the initial condition and move with slope given by the value of f' in [Figure 6.5](#). Different initial conditions lead to different graphs for f , but for a given x -value they all have the same slope (because the value of f' is the same for each). Thus, the different f curves are obtained from one another by a vertical shift. See [Figure 6.6](#).

- Where f' is positive ($1 < x < 3$), we see f is increasing; where f' is negative ($0 < x < 1$ or $3 < x < 4$), we see f is decreasing.
- Where f' is increasing ($0 < x < 2$), we see f is concave up; where f' is decreasing ($2 < x < 4$), we see f is concave down.
- Where $f' = 0$, we see f has a local maximum at $x = 3$ and a local minimum at $x = 1$.
- Where f' has a maximum ($x = 2$), we see f has a point of inflection.

